

MSSC2025 (Turin 09 sept. 2025)
ONE-ELECTRON PROPERTIES



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QM studies

Schrödinger Equation:

$$H\psi = E\psi$$

Approximations: Born-Hoppenheimer
1e Hamiltonian (mean field)

Basis set: Atomic Orbitals:

$$\{\chi\}$$

Secular equation:

$$HC = SCE$$

eigenstates:

$$\psi_q = \sum_p c_{pq} \chi_p$$

Crystalline solids:
Factorisation in reciprocal space, $\mathbf{k}$

Secular equation: $\mathbf{H}(\mathbf{k})\mathbf{C}(\mathbf{k})=\mathbf{S}(\mathbf{k})\mathbf{C}(\mathbf{k})\mathbf{E}(\mathbf{k})$

Basis set: Bloch Functions

$$\{\chi_p(\mathbf{R}_p+\mathbf{r})\} \rightarrow \{\Phi_p(\mathbf{k})\}$$

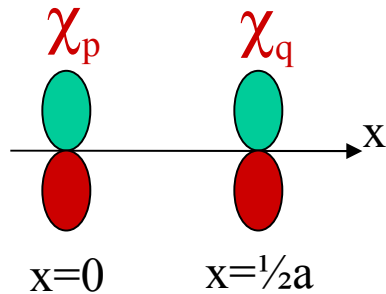
linked by Fourier transform to AOs χ

$$\Phi_p(\mathbf{k}) = 1/\sqrt{N} \sum_{\mathbf{g}} \chi_p(\mathbf{R}_p+\mathbf{r}+\mathbf{g}) e^{i\mathbf{k}(\mathbf{R}_p+\mathbf{g})}$$

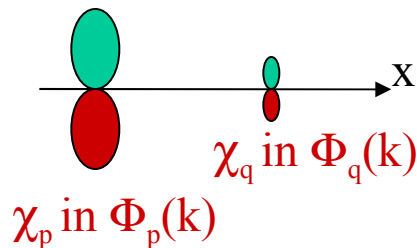
$\Phi_p(\mathbf{k})$ extends over the infinite solid

k-DEPENDENCE IN BLOCH FCTS

$$\Phi_p(k) = \sum_g \chi_p(R_p + r + g) e^{ik(R_p + g)}$$



$$k_x = 0; \quad e^{ikRq} = 1$$



$$0 < k_x < \frac{1}{2}; \quad \text{Re}[e^{ikRq}] < 1$$

In the Bloch fct $\Phi_q(k)$, χ_q appears smaller

Using different k points we highlight the interaction of different AOs in the structure around a reference point

Direct and Reciprocal Spaces (numerical) Fourier transform in each SCF cycle

DIR → REC $H(k)C(k) = S(k)C(k)E(k)$ factorise ∞ problem into a series of finite ones

Calculate eigenvalues/eigenvectors

REC → DIR

DENSITY operator:

Molecule $\rho = \sum_l^{\text{occ}} n_l |\Psi_l\rangle\langle\Psi_l|$

Solid $\rho^g = \int_{\text{IBZ}} \sum_l^{\text{occ}} n_l |\Psi_l(k)\rangle\langle\Psi_l(k)| e^{ikg} dk$

Integral $\int_{\text{IBZ}}$ replaced by finite sum $\sum_{k=1}^n w_k$

Density matrix (solid):

$$P_{pq}^g = \int_{\text{IBZ}} \sum_l^{\text{occ}} n_l c_{pl}(k) c_{ql}^*(k) e^{ikg} dk$$

QM SOLUTION CONTAINS MORE INFORMATION THAN
THE ENERGY. CAN BE EXTRACTED WITH APPROPRIATE
ANALYSIS METHODS

Direct and reciprocal spaces



Analysis of the Quantum Mechanical
solution for solids can be done in both spaces

DIRECT SPACE: 1-electron density $\rho(\mathbf{r})$
similar to molecular systems

RECIPROCAL SPACE: eigenvalues $\epsilon_1(\mathbf{k})$
unique to solids

UCL

Analysis of the electron density $\rho(\mathbf{r})$

DIRECT SPACE

-VISUAL: 2D and 3D plots (**ECHG, ECH3**)

-NUMERICAL: population analysis

Mulliken (**PPAN**)

Hirshfeld

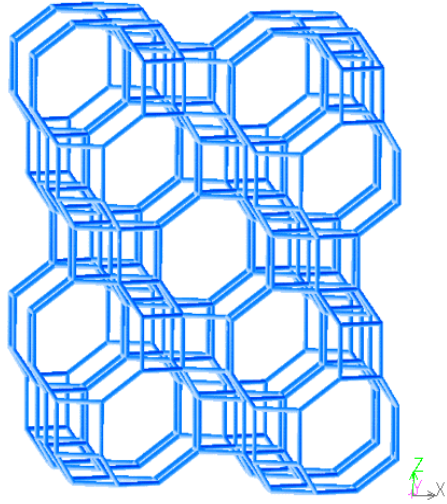
-TOPOLOGICAL analysis: BADER's AIM

-WANNIER functions; natural orbitals

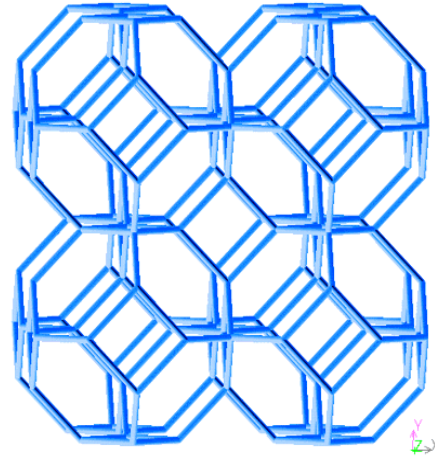
**TIP: ALWAYS CHOOSE A SUITABLE
REFERENCE TO COMPARE WITH!**

EXAMPLE: FRAMEWORK PROPERTIES OF ZEOLITES (SiO_2) AND ALPOs (AlPO_4)

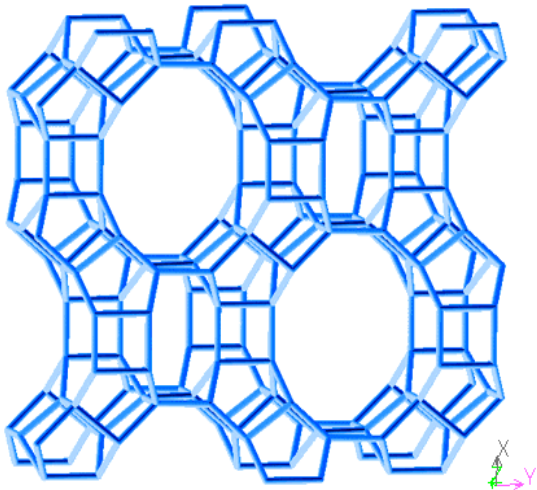
CHA



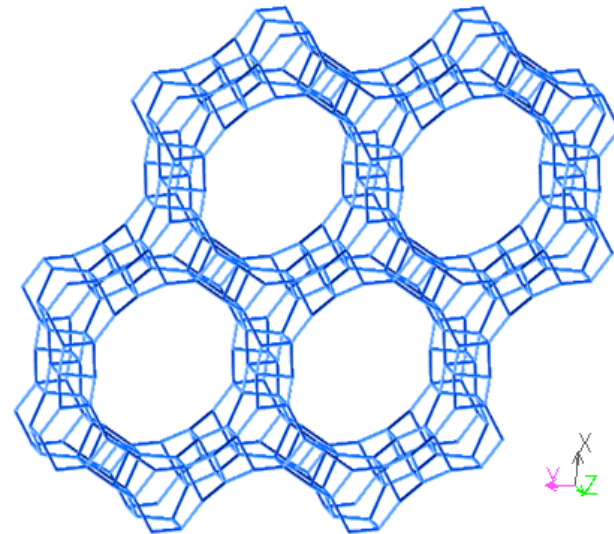
SOD



MOR



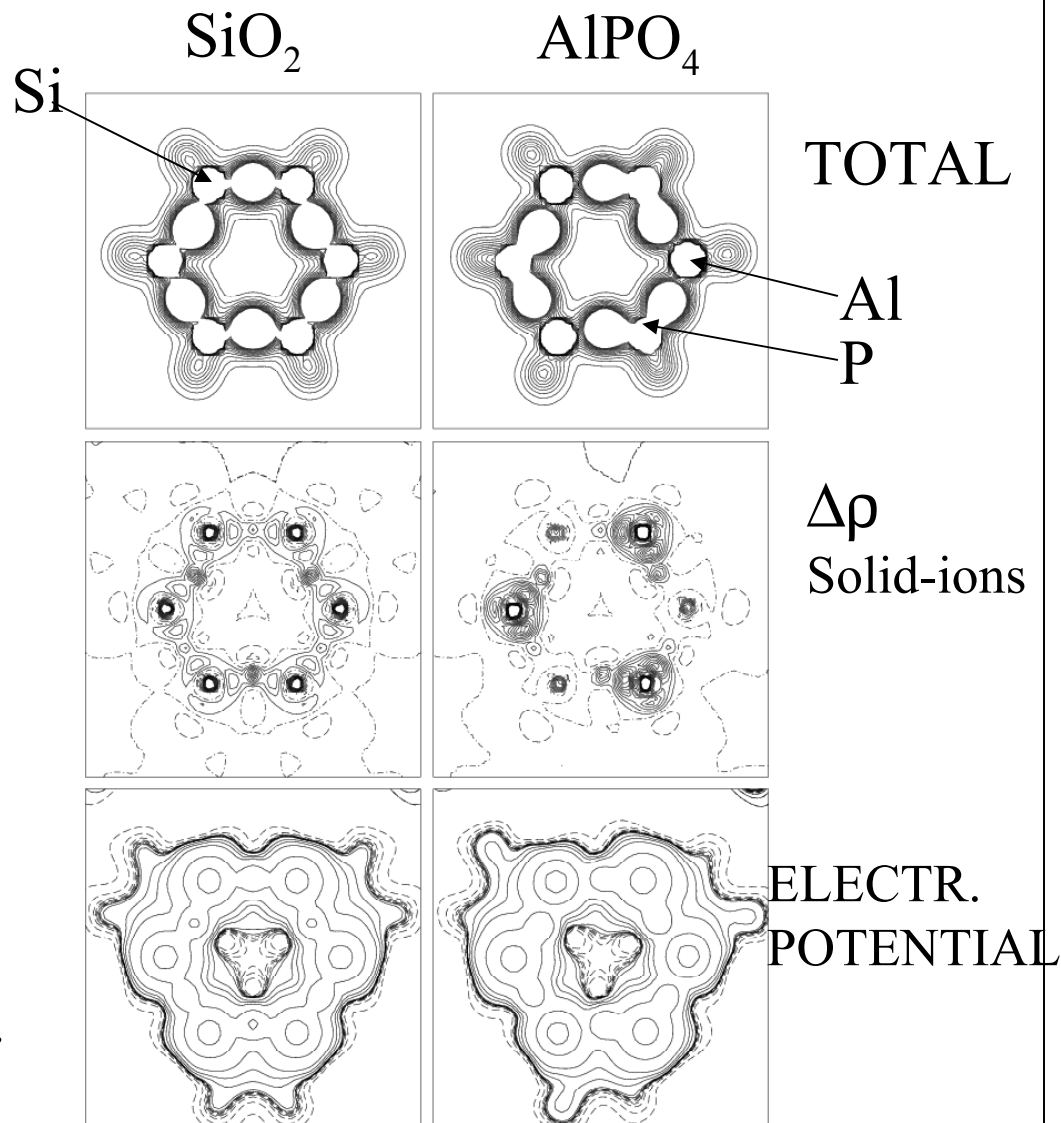
VFI



Visual analysis
of $\rho(r)$
2D PROJECTION

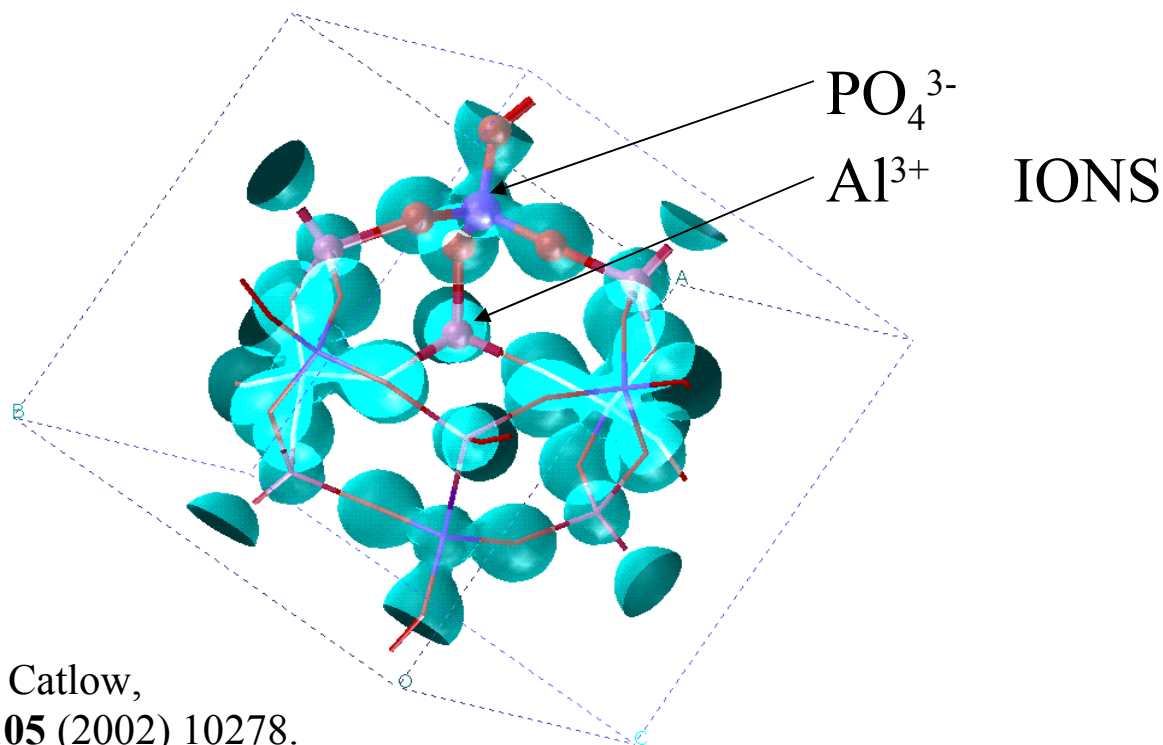
Framework properties of
zeolites and AlPOs

- 1) SELECT A
REPRESENTATIVE PLANE
- 2) SUITABLE REFERENCE:
ISOLATED FORMAL IONS
(SPHERICAL SYMMETRY).



Visual analysis of $\rho(r)$ 3D PROJECTION

Framework properties of AlPOs
ISODENSITY LEVEL

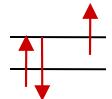
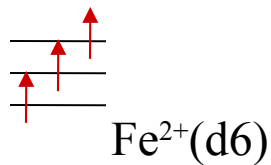


F. Corà and C.R.A. Catlow,
J. Phys. Chem., **B 105** (2002) 10278.

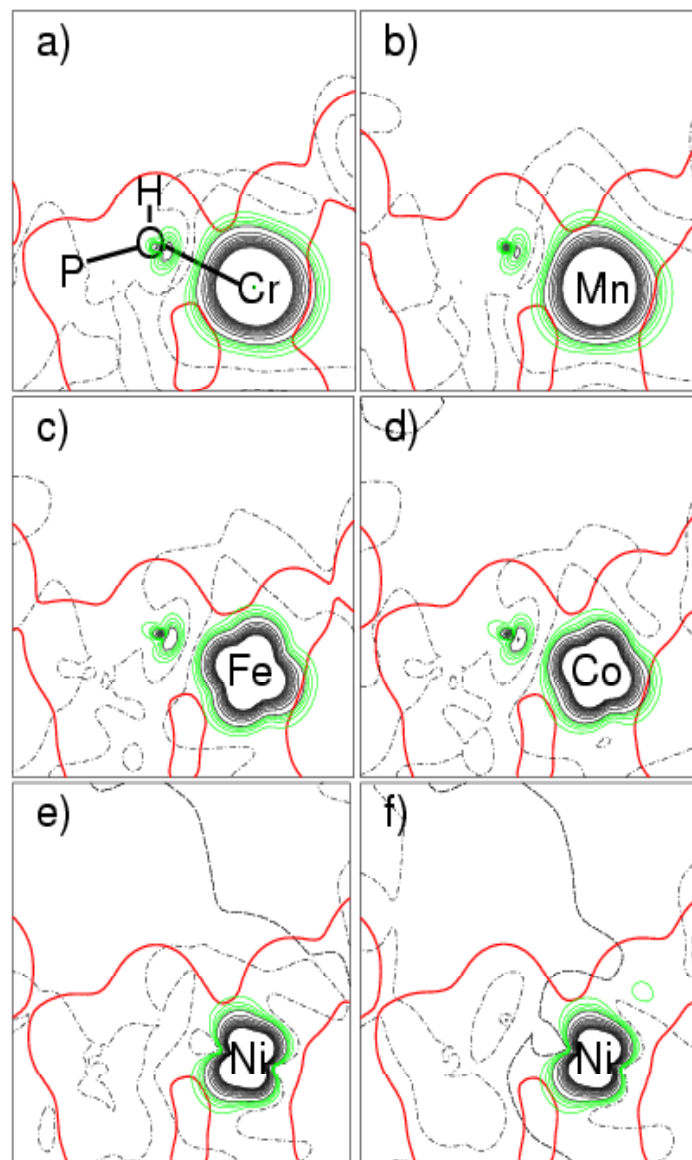
LEWIS ACIDITY of TRANSITION METAL DOPED ALUMINOPHOSPHATES

spin density ($\rho^\alpha - \rho^\beta$) of the TM
ion
compared to the
framework size

SUITABLE REFERENCE:
MINORITY SPIN DENSITY



F. Corà, I Saadoune and C.R.A. Catlow,
Angew. Chemie, **41** (2002) 4871.



NUMERICAL analysis of the Electronic density $\rho(\mathbf{r})$

MULLIKEN definition of charges (PPAN)
representation of $\rho(\mathbf{r})$ in the basis set $\{\chi_p\}$ chosen

$$M_{pq}^g = P_{pq}^g \cdot S_{pq}^g \quad \text{Mulliken matrix}$$

ORBITAL charge: $q_p = \sum_{g,q} M_{pq}^g$

ATOMIC charge: $q_A = \sum_{p \in A} q_p$

BOND charge: $q_{AB} = 2 \sum_{p \in A, q \in B} M_{pq}$

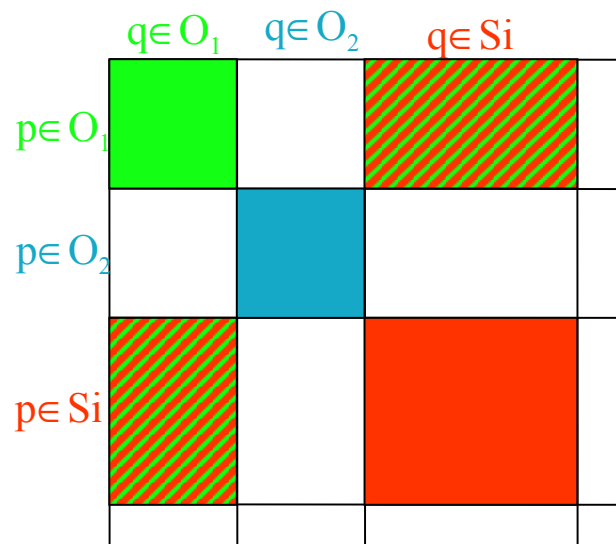
MULLIKEN ANALYSIS (local AO basis set)

Each atom A is associated an 'atomic' basis set $\{\chi_{p \in A}\}$

Define the matrix $M^g = P^g \cdot S^g$ ($M_{pq}^g = P_{pq}^g \cdot S_{pq}^g$)

Attribute the charge $P_{p \in A, q \in B}$ to the atoms A,B

Example: SiO_2



Out-of-diagonal terms: bond population $q_b(O_1-Si)$

Diagonal terms: attributed to 1 atom $q(O_1)$

Atomic charges: $Q(A) = q(A) + \frac{1}{2} \sum_{B \neq A} q_b(A-B)$; orbital charge: $q_p = \sum_{q,g} M_{pq}^g$

	T	Q(T)	Q(O)	$q_b(T-O)$
SiO_2	Si	+2.08	-1.04	0.29
AlPO_4	P	+2.85	-1.26	0.27
	Al	+2.19	-1.26	0.14
GaPO_4	P	+2.87	-1.30	0.27
	Ga	+2.33	-1.30	0.11

UCL

NUMERICAL analysis of the Electronic density $\rho(\mathbf{r})$

HIRSHFELD definition of charges
comparison of $\rho(\mathbf{r})$ with atomic references

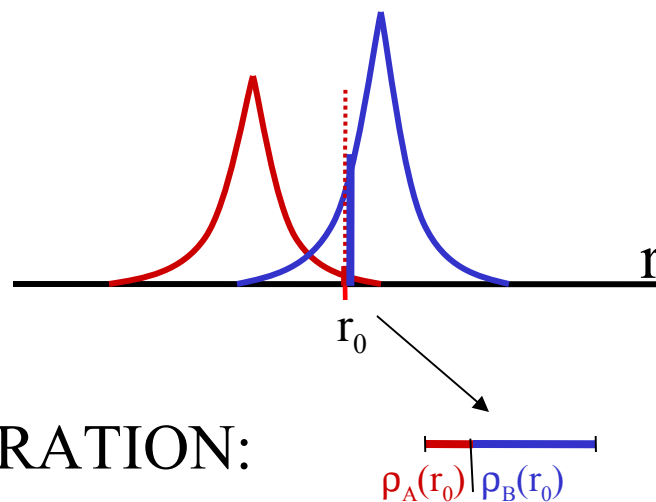
ATOMIC WEIGHTS: $W_A(\mathbf{r})$

$$W_A(\mathbf{r}) = \rho_A(\mathbf{r}) [\sum_B \rho_B(\mathbf{r})]^{-1}$$

ATOMIC CHARGES BY INTEGRATION:

$$q_A = \int \rho_d(\mathbf{r}) W_A(\mathbf{r}) d\mathbf{r}$$

$$\rho_d(\mathbf{r}) = \rho(\mathbf{r}) - \sum_A \rho_A(\mathbf{r})$$



BADER ANALYSIS

Calculate the Laplacian of $\rho(\mathbf{r})$: $L = \nabla^2 \rho(\mathbf{r})$

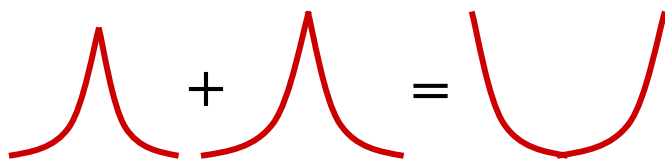
in zones of covalent interaction: $L < 0$

in zones of ionic interaction: $L > 0$

Electron-density profile along bonds

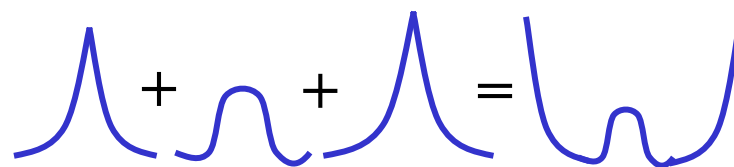
IONIC interaction

COVALENT interaction



Curvature > 0

$L > 0$



Curvature < 0

$L < 0$

TOPOND, by C. Gatti, CNR Milano

BADER ANALYSIS

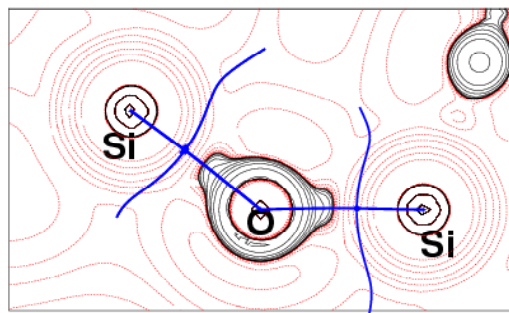
Framework properties of
zeolites and AlPOs

Laplacian $L = \nabla^2 \rho(r)$

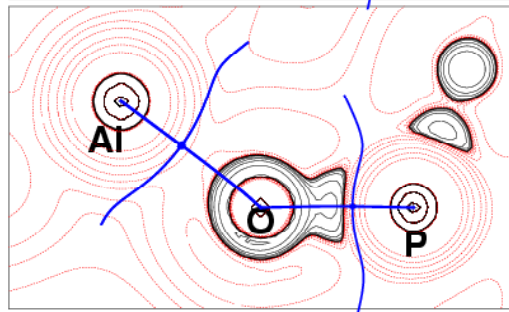
RED: $L > 0$ zones of ionic
interaction in the solid

BLACK: $L < 0$ zones of covalent
interaction in the solid

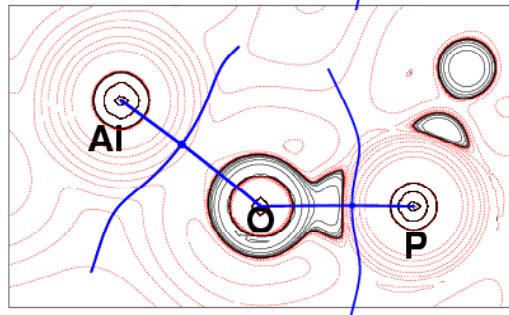
BLUE: molecular graphs
delimit atomic basins



Si-O-Si bond, silica
Hartree-Fock density



Al-O-P bond, AlPO
Hartree-Fock density

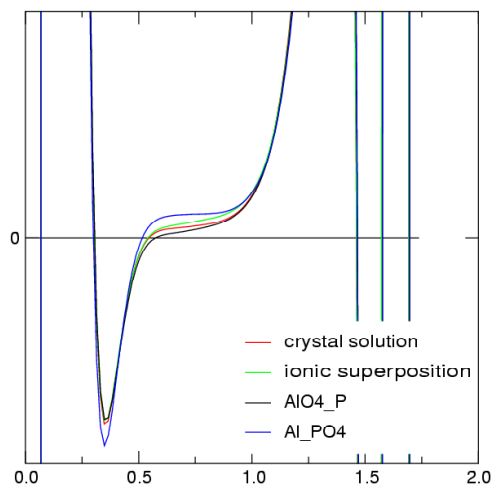


Al-O-P bond, AlPO
DFT-GGA density

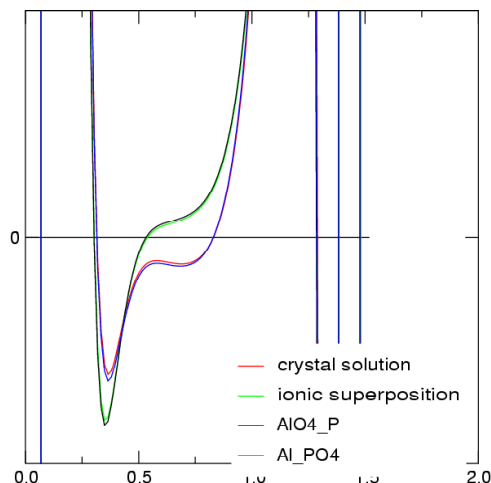
F. Corà and C.R.A. Catlow,
J. Phys. Chem., **B 105** (2002) 10278.

BADER ANALYSIS

$\nabla^2\rho(r)$
along
Al-O
bond



$\nabla^2\rho(r)$
along
P-O
bond



Framework properties of AlPOs

LAPLACIAN $L=\nabla^2\rho(r)$

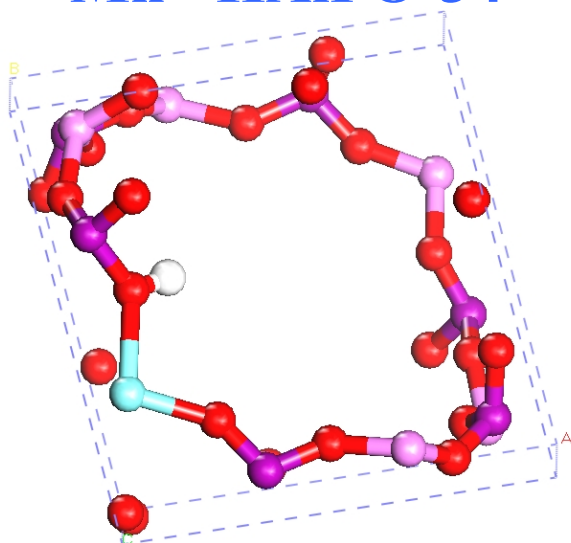
$L>0$ zones of ionic
interaction in the solid

$L<0$ zones of covalent
interaction in the solid

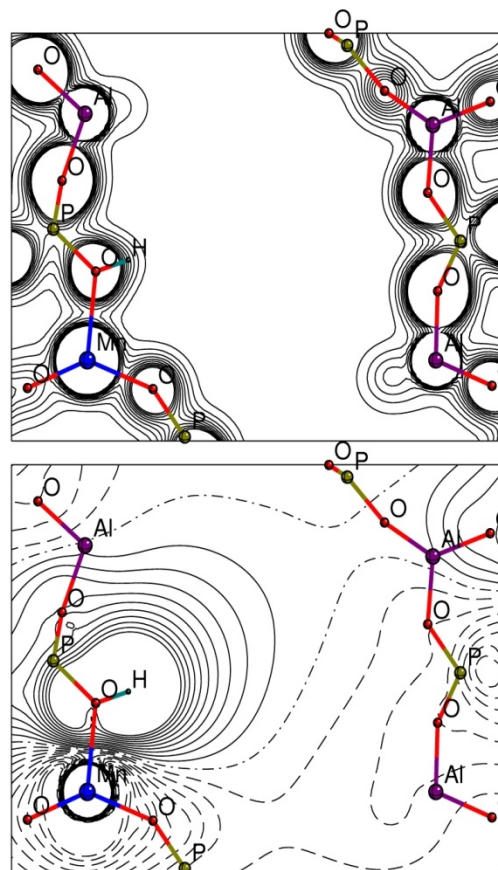
REFERENCE: ISOLATED IONS
OR AlO_4/PO_4 GROUPS

(DIFFERENCE) ELECTROSTATIC POTENTIAL

Mn^{II}-HAIPO-34



SUITABLE REFERENCE:
UNDOPED SYSTEM
 $\Delta V(\text{MnAlPO}-\text{AlPO})$



I.Saadoune, C.R.A.Catlow, K.Doll,
F.Corà, Molec. Simul., 30 (2004) 607.

OXIDATION OF HYDROCARBONS WITH O₂ IN MeAlPO-5 (Mn, Fe, Co)



Bond-breaking is an electronic process

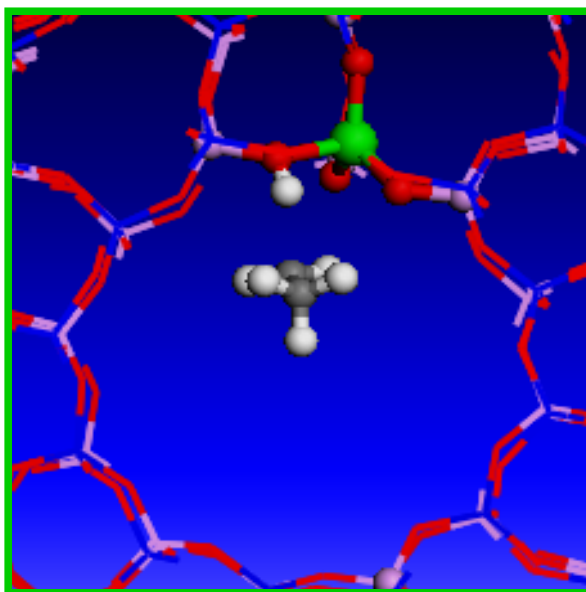
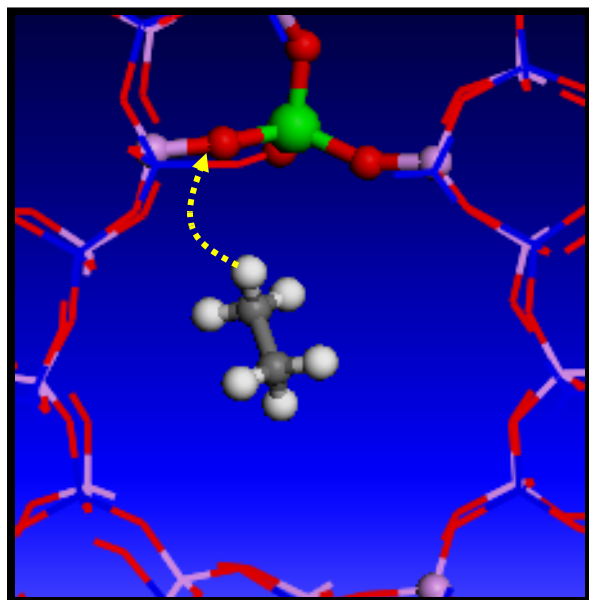
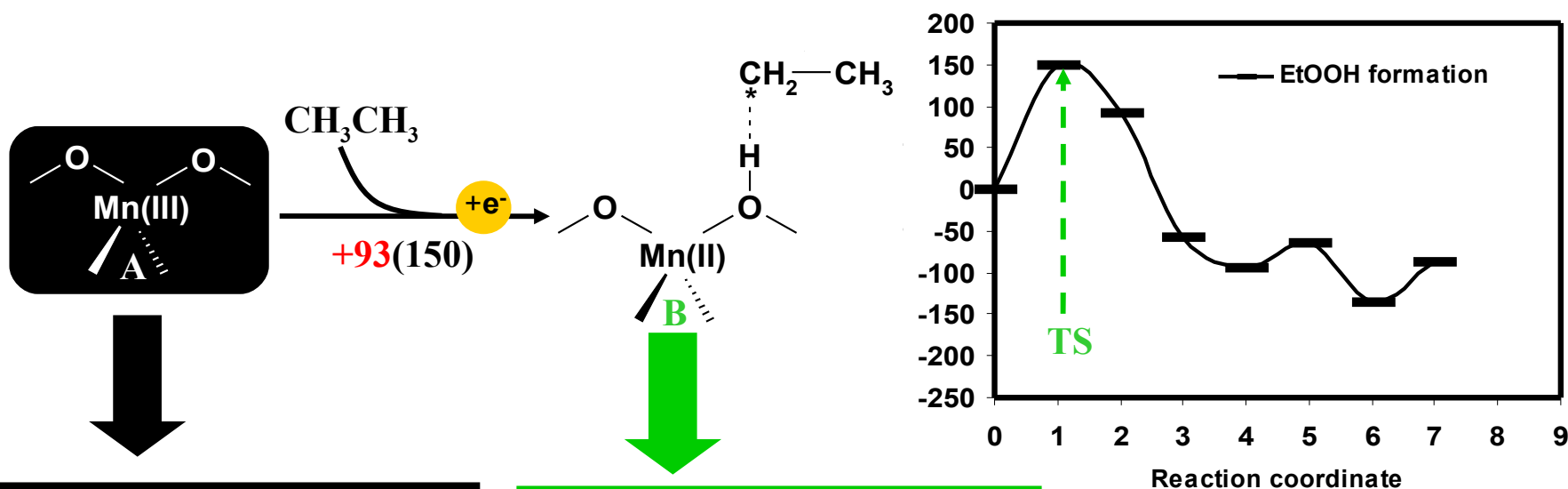
We need to locate transition states, not only minima

Use ethane as model hydrocarbon

TS search under PBC

Hybrid-exchange functionals within DFT

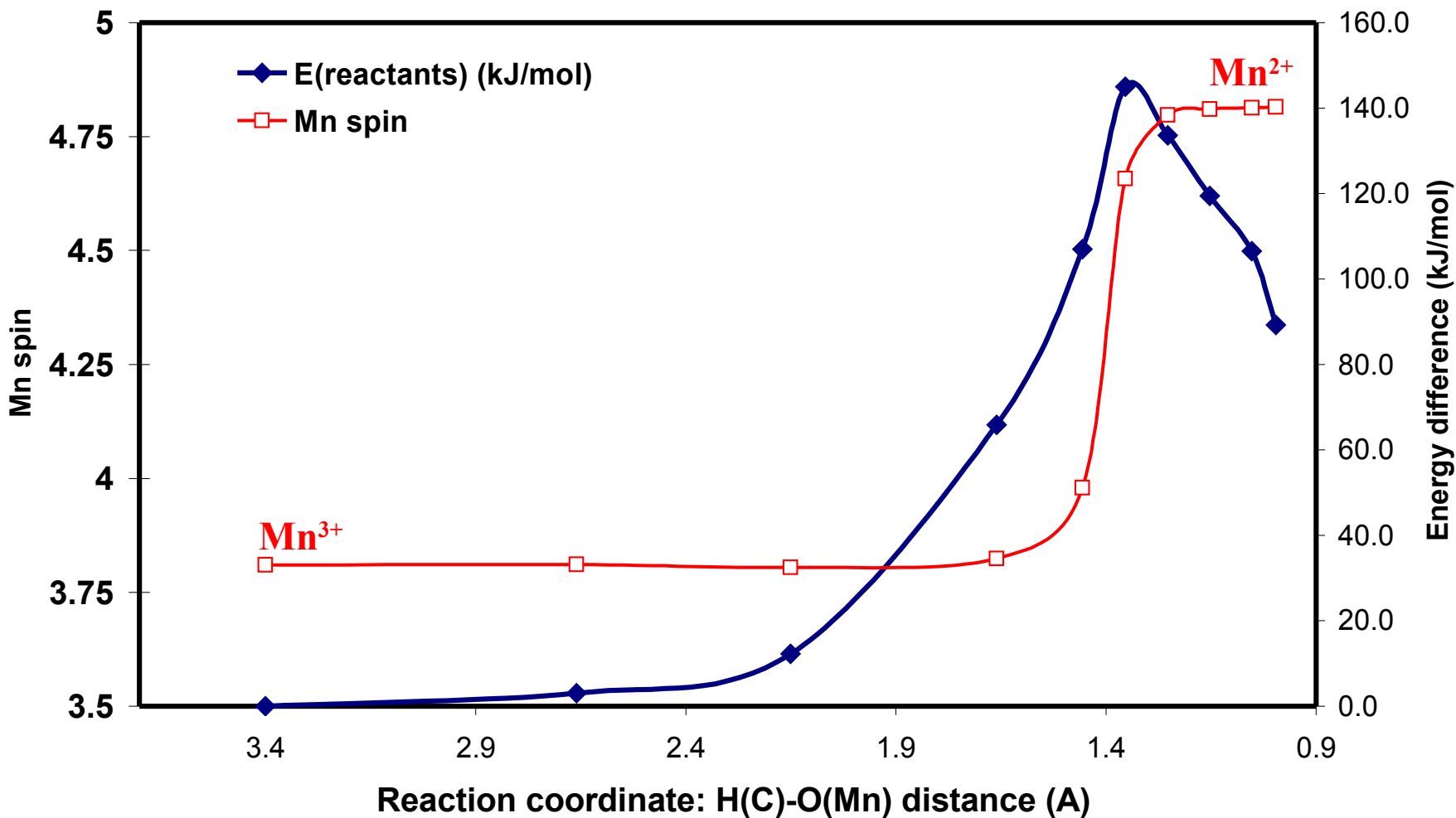
The crucial step: ethane activation (radical mechanism)



L. Gomez-Hortiguela,
G. Sankar, C.R.A. Catlow,
F. Cora, Chem Eur J

C-H bond activation

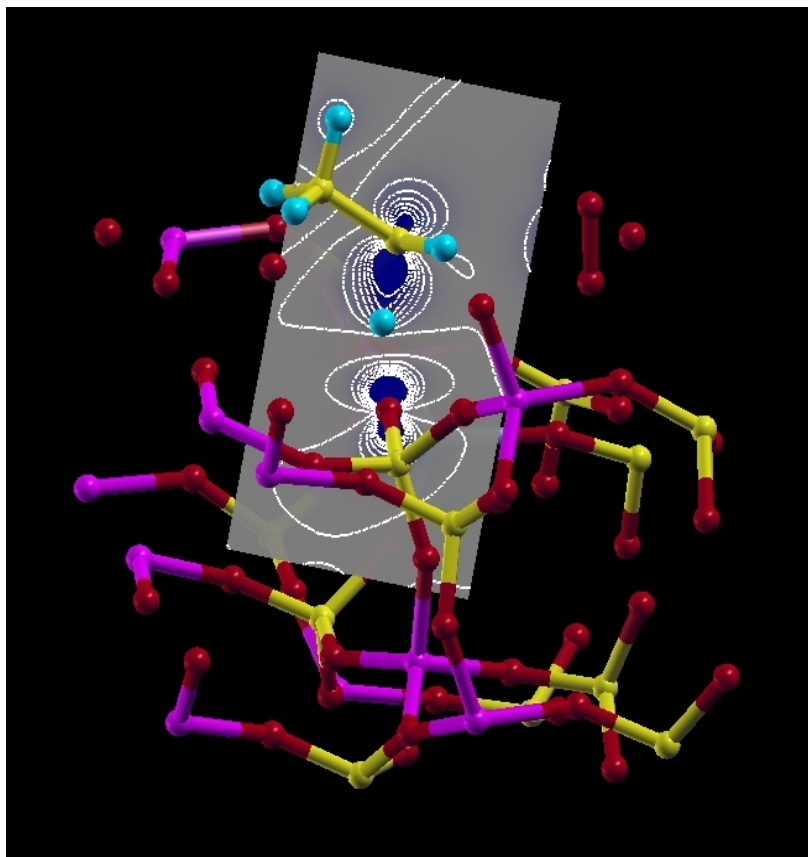
Activation of the C-H bond: reduction of Mn



-Mn is reduced from (III) to (II): driving force of this reaction

C-H bond activation: the transition state

In homolytic reactions, unpaired e- delocalised in TS



Crucial to go beyond standard DFT.

Systematic errors for localised electronic states due to SELF INTERACTION.

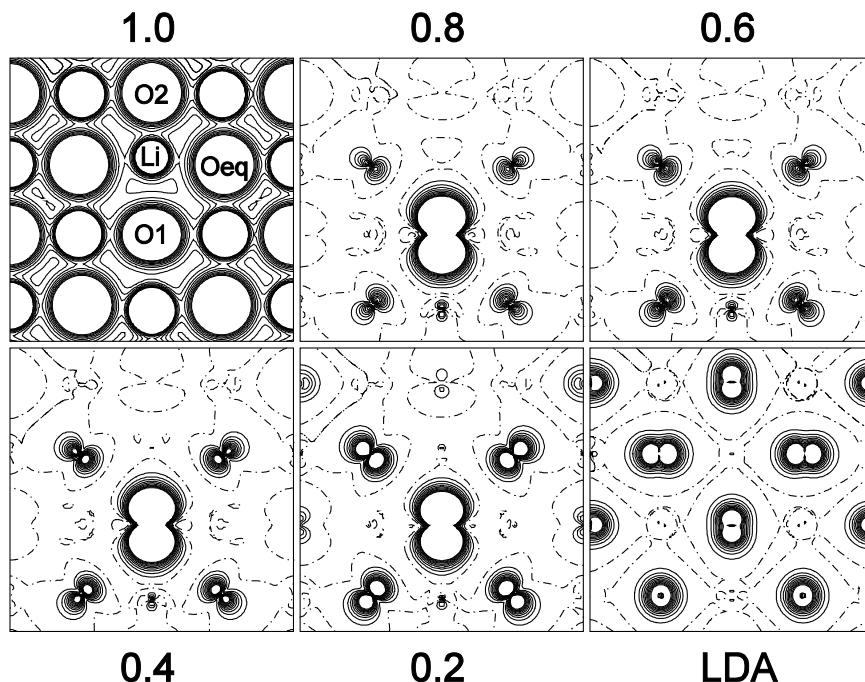
Hybrid exchange required to correct errors.

Spin density map

SPIN DENSITY

Li-hole in MgO

Comparison of Hamiltonians -Hybrid exchange functionals



TOTAL ($a_x=1.0$)
AND
SPIN ELECTRON
DENSITY

F. Corà et al., page 171-232 in J. McGrady and N. Kaltsoyannis (Ed), DFT in Inorganic Chemistry, Structure and Bonding, Vol. 113, Springer-Verlag, Heidelberg, 2004.

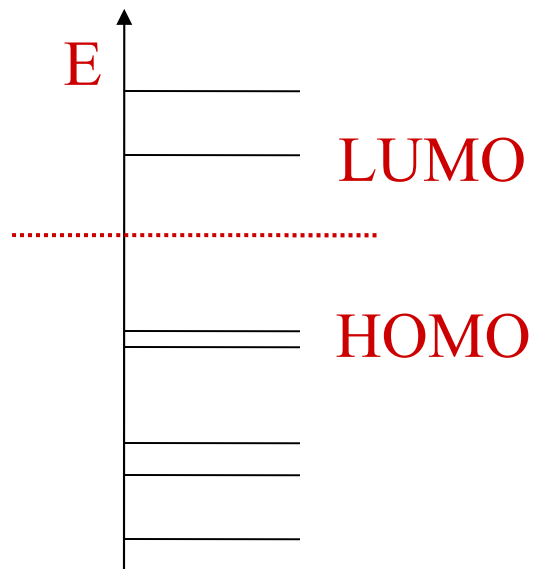
UCL

Reciprocal space properties
K-dependence of the solution

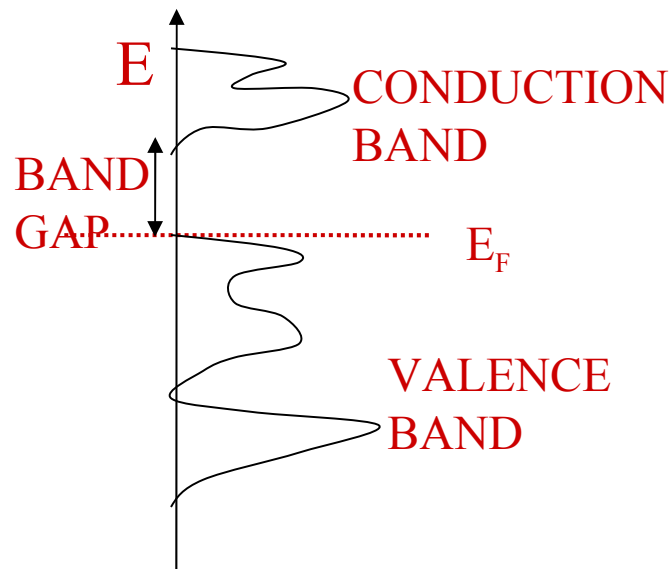
$$H(k)C(k) = S(k)C(k)E(k)$$

Spectrum of one-electron energy levels

Molecule

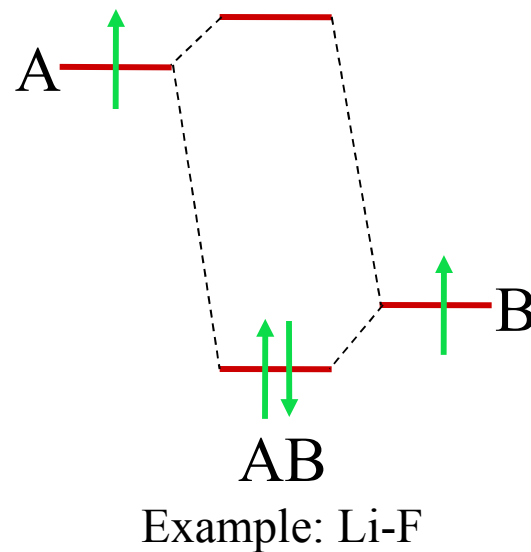
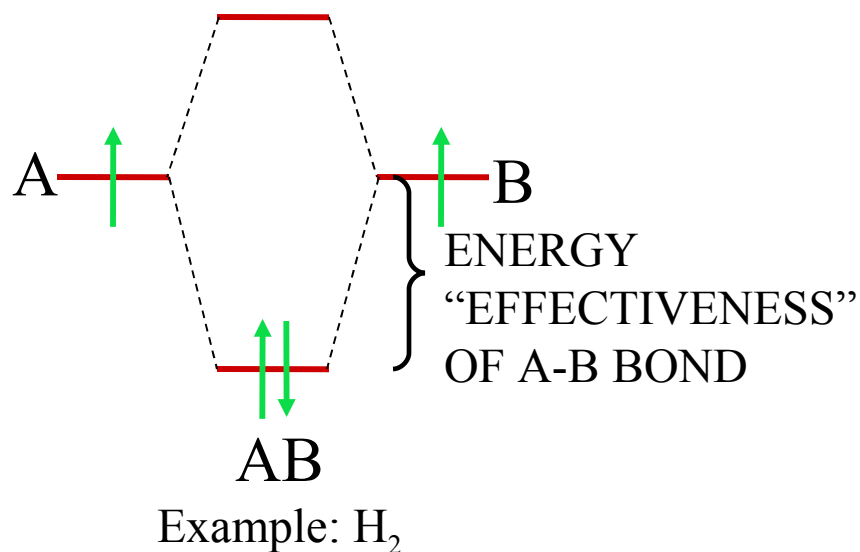


Solid



CHEMISTRY FROM ENERGY LEVELS

reference to MO diagram formolecules



Energetic effectiveness of covalence in the A-B bond

How similar are the MO to the starting AO's?

(AO=Atomic or ionic orbitals)

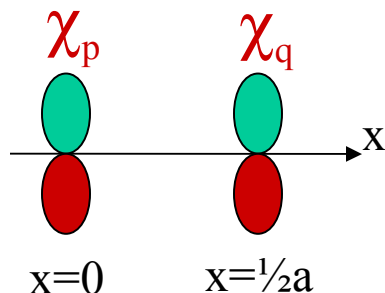
Equivalent information for a solid? BAND STRUCTURE

CHEMISTRY FROM ENERGY LEVELS

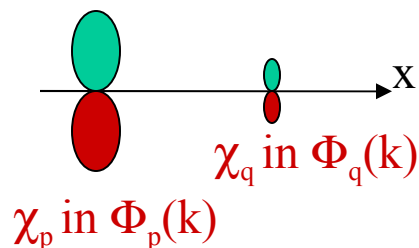
exploit the k -dependence of the Bloch functions

$$H(k)C(k)=S(k)C(k)E(k); H_{pq}(k)=\langle \Phi_p(k)|H|\Phi_q(k)\rangle$$

$$\Phi_p(k) = \sum_g \chi_p(R_p+r+g) e^{ik(R_p+g)}$$

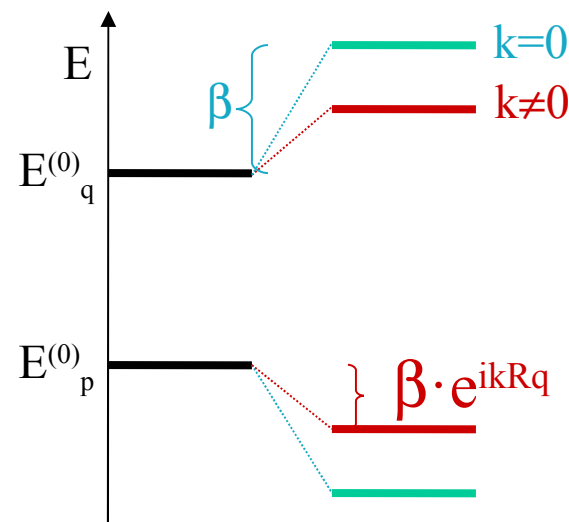


$$k_x = 0; e^{ikRq} = 1$$



$$0 < k_x < 1/2; \text{Re}[e^{ikRq}] < 1$$

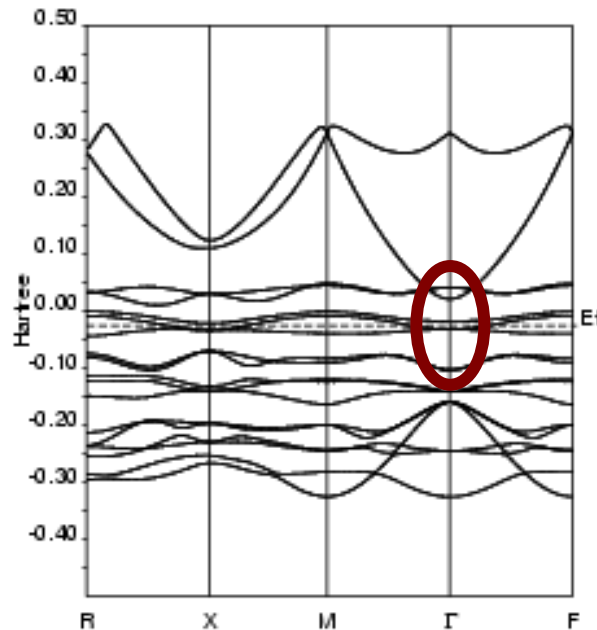
In the Bloch fct $\Phi_q(k)$, χ_q appears smaller



Result: dispersion of eigenvalues in k -space

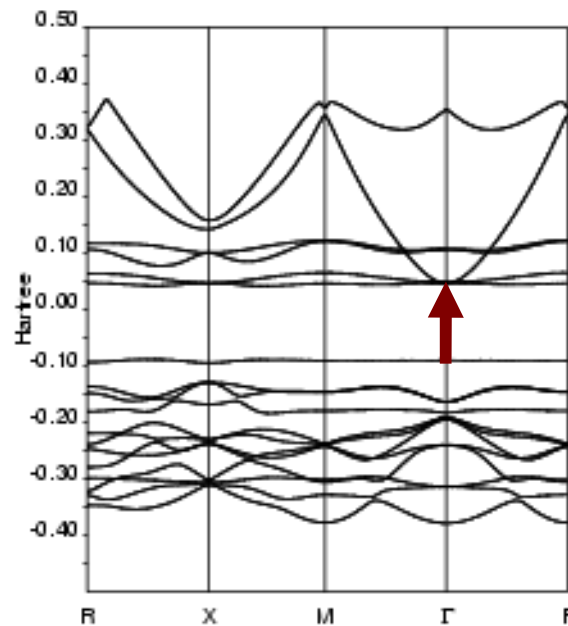
Band Structure of FeO

0%HF



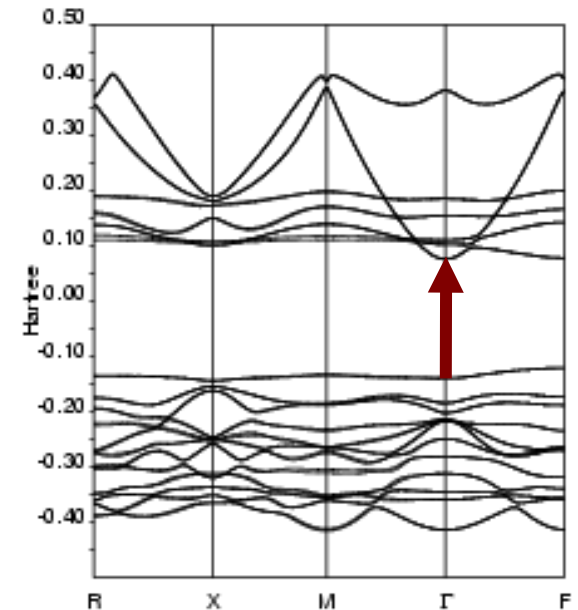
BLYP functional gives
a metallic solution

20%HF



B3LYP transition between
 $\text{Fe}3d/\text{O}2sp - \text{Fe}3d$

35%HF



H35B65LYP
transition between
 $\text{Fe}3d/\text{O}2sp - \text{Fe}4s$

Experimentally weak absorption of 0.5 eV between $\text{Fe}3d/\text{O}2sp - \text{Fe}4s$
and stronger transition of 2.4 eV between $\text{Fe}3d/\text{O}2sp - \text{Fe}3d$

Bagus et al. PRL (1977)

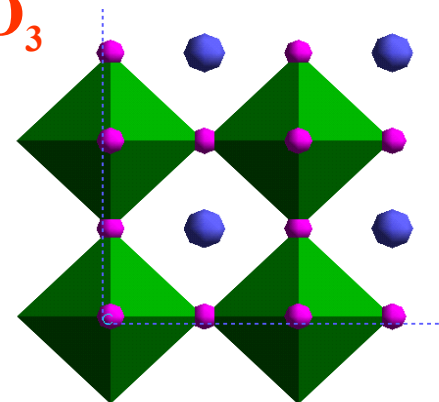
Alfredsson et al. PRB (2004)

PEROVSKITE STRUCTURE: AMO_3

For more informaton:

F. Corà and C.R.A.Catlow, Faraday Discussions 114 (1999) 421

F. Corà et al., '*The performance of hybrid density functionals in solid state chemistry*', Structure and Bonding, Vol. 113, page 171-232



ALL IONS (M,O) TOPOLOGICALLY EQUIVALENT

Network of corner-sharing MO_6 octahedra

Dodecahedral interstices A (can be empty)

2 cationic sites: different partition of charge possible

LaMnO_3 (3-3)

BaTiO_3 (2-4)

KNbO_3 (1-5)

WO_3 (0-6)

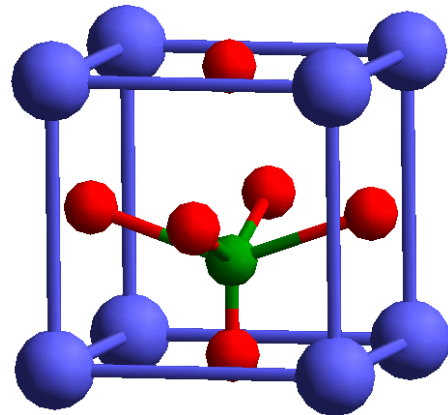
Does this affect the solution?

FERROELECTRIC PEROVSKITES

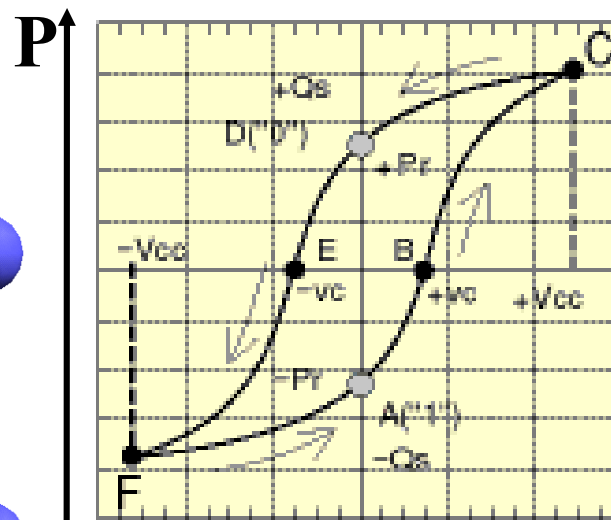


FERROELECTRIC DISTORTION IN $d^{(0)}$ PEROVSKITES

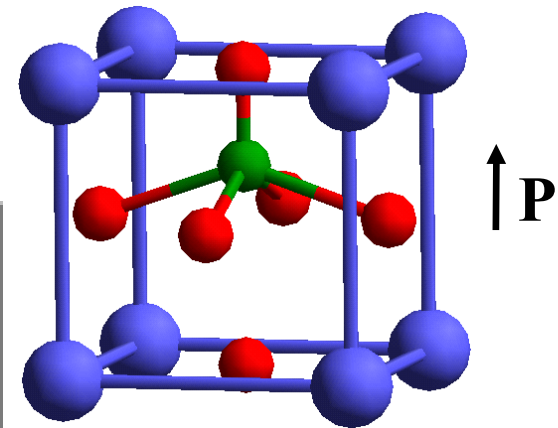
M OFF-CENTERING



$P \downarrow$



Hysteresis loop



$P \uparrow$

LABELLING

UP-DOWN

ON-OFF ATOMIC SWITCH

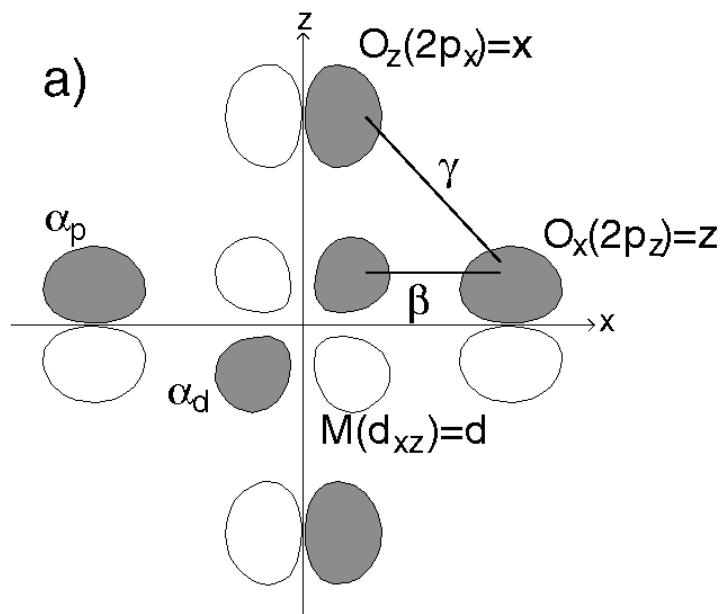
0-1 BIT IN COMPUTING

(FE-RAM)

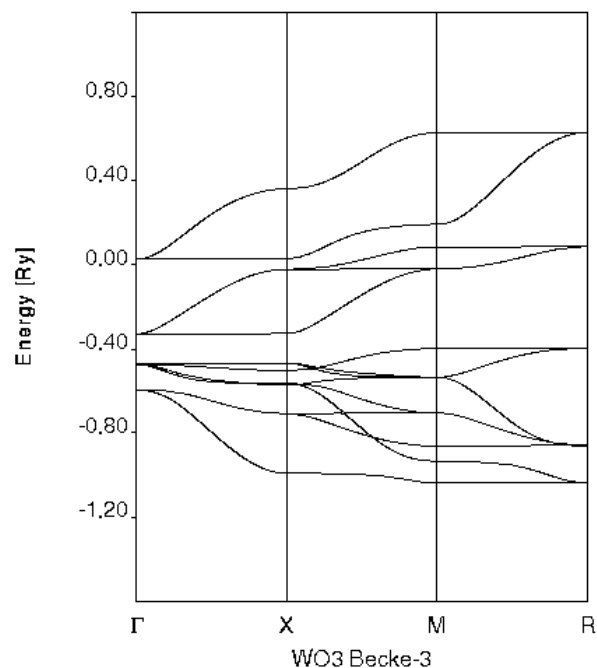
Applied voltage

FEATURES OF THE QM SOLUTION – CUBIC PHASE

FRONTIER AO's



BAND STRUCTURE



Solution modulated by a wave e^{ikR}
 Different in different k points

CUBIC PHASE -A COMPARISON OF DIFFERENT PEROVSKITES

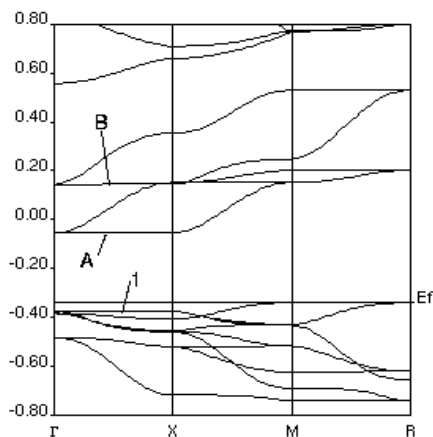
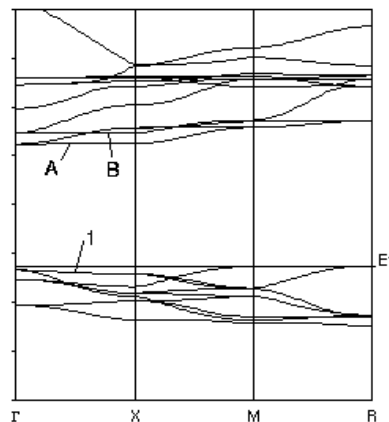
CHARGE
PARTITION

(6-0)

(5-1)

(4-2)

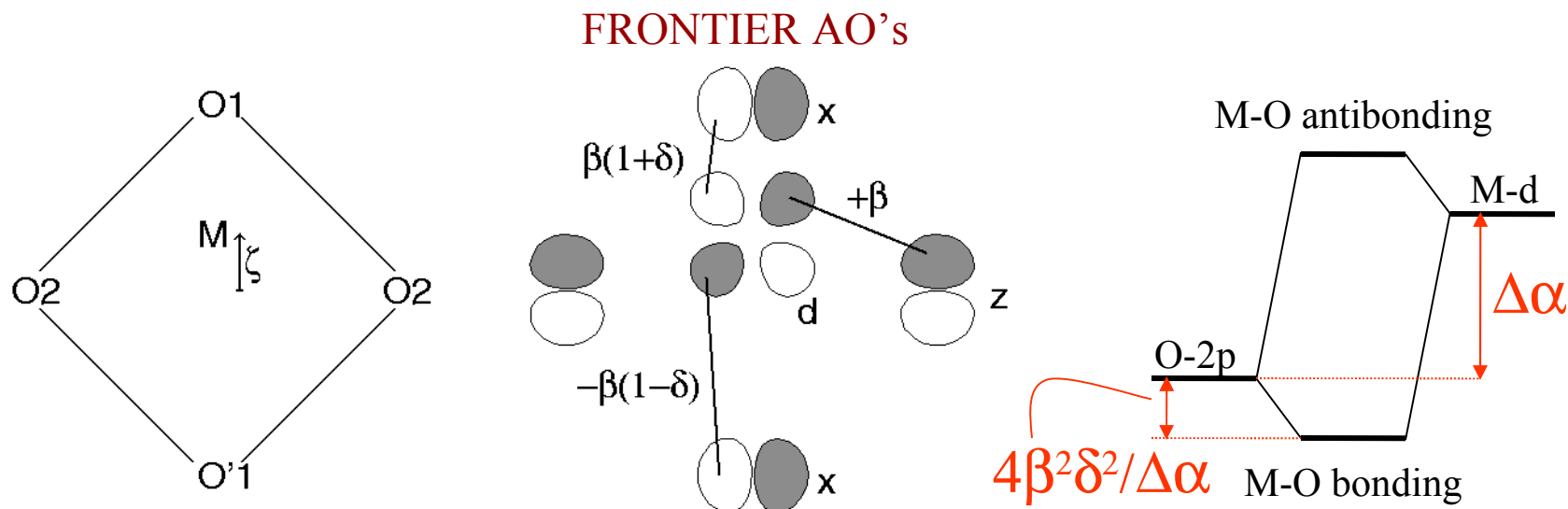
Material	a	Q(O)	$\Delta\alpha$	β	γ^*10	$\beta^2/\Delta\alpha$
MoO ₃	3.78	-0.998	0.299	0.126	-0.162	0.053
WO ₃	3.77	-1.362	0.317	0.130	-0.168	0.053
LiNbO ₃	3.96	-1.385	0.412	0.116	-0.161	0.033
NaNbO ₃	3.98	-1.374	0.399	0.114	-0.157	0.032
KNbO ₃	4.03	-1.395	0.391	0.108	-0.151	0.030
CaTiO ₃	3.88	-1.582	0.514	0.120	-0.204	0.028
SrTiO ₃	3.93	-1.609	0.505	0.116	-0.195	0.027
BaTiO ₃	4.02	-1.571	0.474	0.101	-0.188	0.022

WO₃CaTiO₃

THE HIGHER THE FORMAL
CHARGE OF M,
THE LOWER THE BAND GAP,
THE HIGHER COVALENCE

SUITABLE REFERENCE:
SYSTEM WITH EXTREME
BEHAVIOUR

FEATURES OF THE DISTORTED (FE) PHASE



M-O COVALENCE NO LONGER SYMMETRY-FORBIDDEN

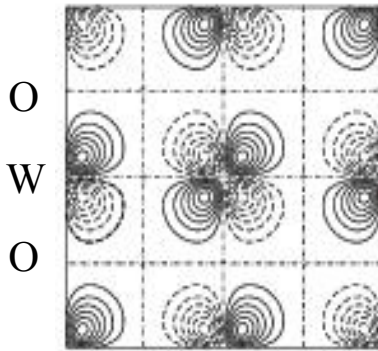
M-O NON BONDING LEVELS HYBRIDISE INTO BONDING/ANTIBONDING PAIR

EFFECTIVENESS PROPORTIONAL TO $\beta^2/\Delta\alpha$ AND M OFF-CENTERING

SAME TREND WITH FORMAL CHARGE OF M (see cubic phase) APPLIES HERE

FERROELECTRIC DISTORTION IN PEROVSKITES

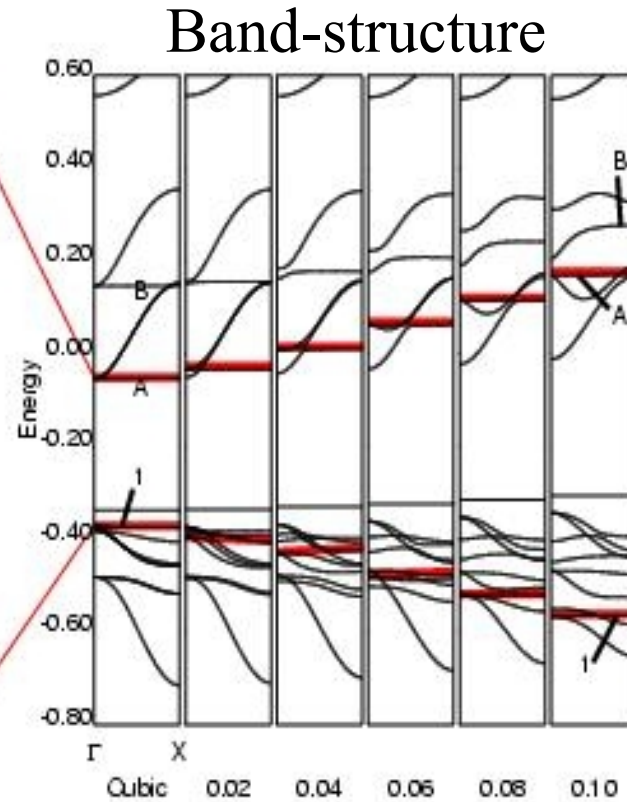
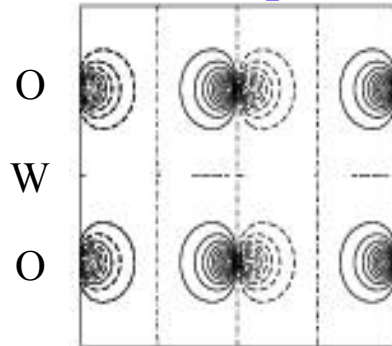
The case of WO_3



Pure M-d level

Wave-function in Γ

Pure O-2p level

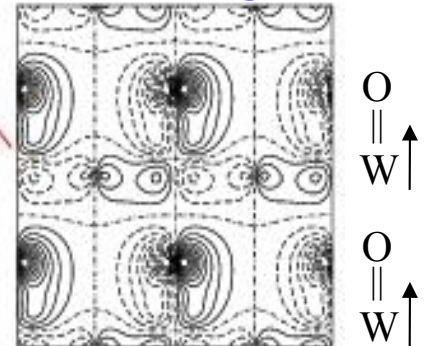


W off-centering



M-O antibonding level

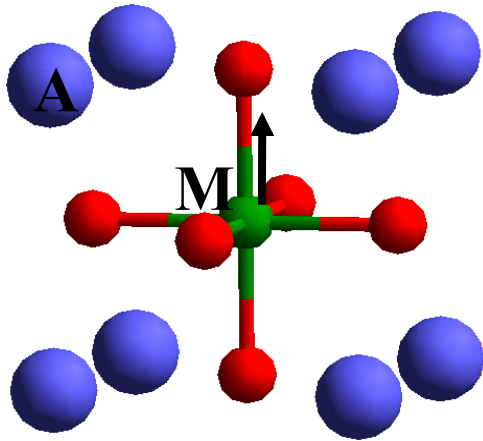
M-O bonding level



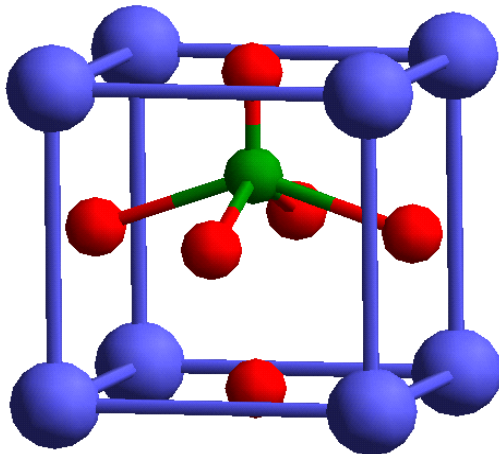
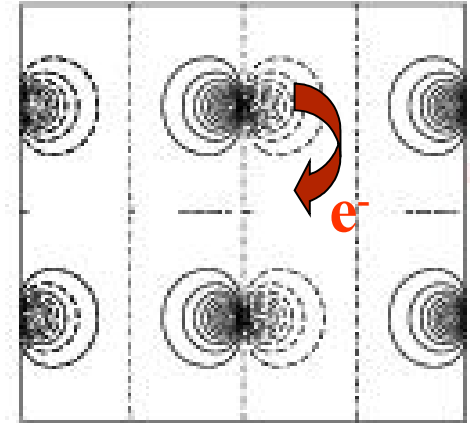
Cubic phase: M-O interactions are symmetry-forbidden

FE distortion forms M-O bonding-antibonding pair

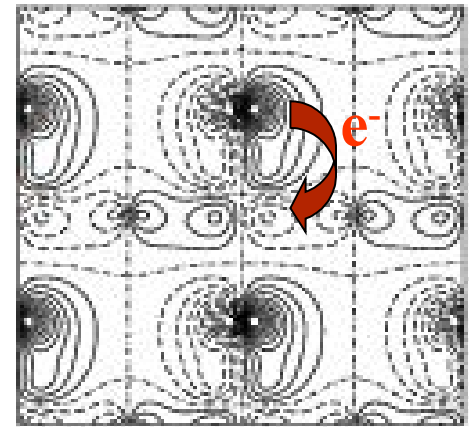
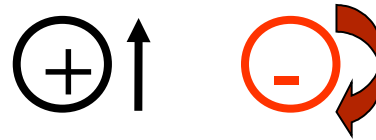
DISPLACEMENT OF CHARGES DURING FERROELECTRIC DISTORTION



CUBIC PHASE
 $2e^-$ in O-2p AO



FE PHASE
 $2e^-$ in MO BOND



THE POLARISATION HAS
NUCLEAR AND ELECTRONIC CONTRIBUTIONS

UCL

DOS, PDOS (integrals of band structure over k)
How many levels available at each energy, $N(e)$

Based on Mulliken partition of charges

ORBITAL PDOS:

$$N_p(e) = 1/V_B \int_{IBZ} \sum_{q,l,g} n_l(k) S_{pq}(k) c_{pl}(k) c_{ql}^*(k) e^{ikg} \delta[e - e_l(k)] dk$$

ATOMIC PDOS:

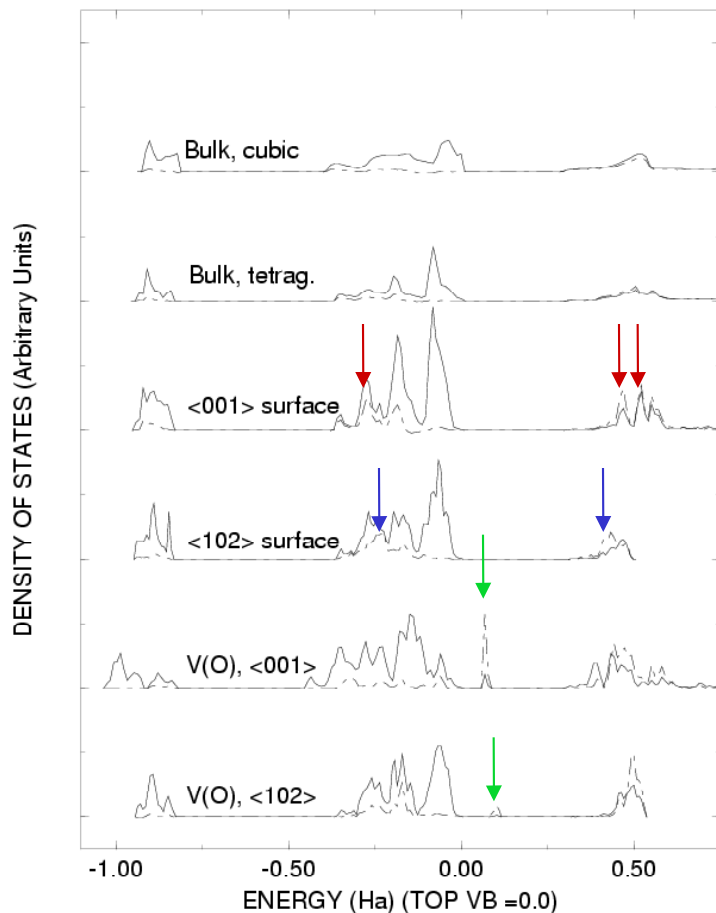
$$N_A(e) = \sum_{p \in A} N_p(e)$$

TOTAL DOS:

$$N_{\text{tot}}(e) = \sum_A N_A(e)$$

In the integration we lose information; DOS less informative of the band structures, but simpler for big systems.

DOS, PDOS (How many levels available at each energy, $N(e)$)



WO_3 : bulk, surfaces, defects

Surface states

kink states

defect states

~200 atoms; impossible to
Highlight in band-structure.

- **Many ways to analyse the solution**
- **Add value to publications at little additional cost**
- **Not a single recipe: depends on what we need to prove**

- **Often relies on model descriptions, textbook concepts**
- **Always useful to compare with reference state**
fully understood: e.g. perfect ionic state, closed shell density
highlight differences in real system

- **Complementary information in direct (electrons) and reciprocal space (energy levels)**